Practical - 8

Practical Title : MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements

Problem Definition: Design the database for StudentHub with students, events and registrations. Create PHP database connection using MYSQL statements.

Theory

1. **Is the schema normalized properly?**
   Answer: Yes, the schema is properly normalized to reduce data duplication and maintain data consistency.

2. **Are primary keys and foreign keys correctly defined?**
   Answer: Yes, primary keys uniquely identify records, and foreign keys correctly establish relationships between related tables.

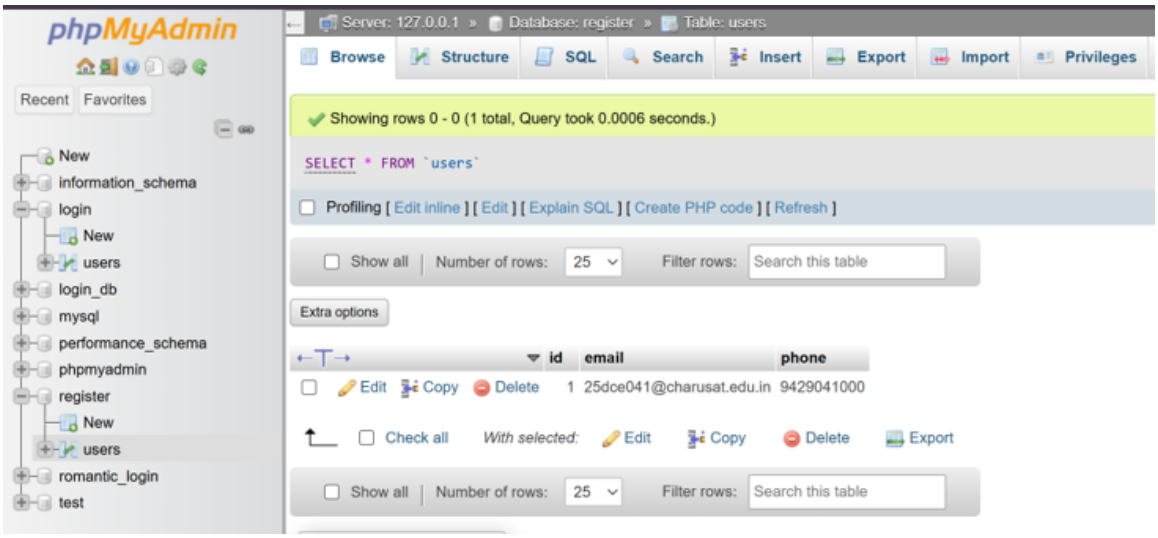
3. **Is the database connection handled securely?**
   Answer: The database connection should use secure credentials and prepared statements to protect against unauthorized access and SQL injection.

Implementation :

- Created the database and required tables.
- Defined primary keys for unique record identification.
- Added foreign keys to establish relationships between tables.
- Applied proper data types and constraints.
- Implemented secure database connection.
- Used validation and prepared statements for safer data handling.

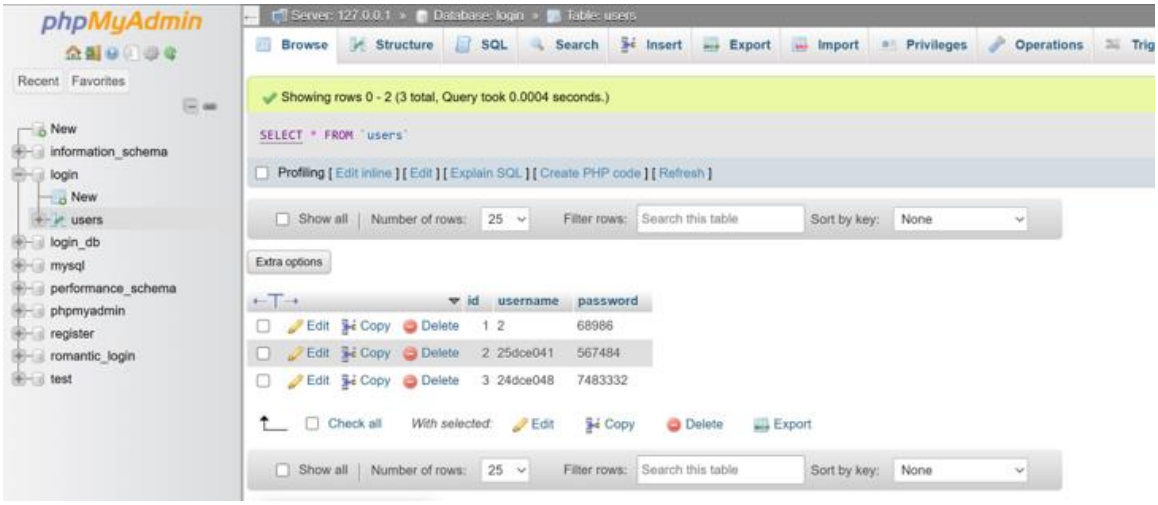
A. Evidence of Work (Screenshots)

B.



C.

D.



E.

F. GitHub Repository Link:

https://github.com/25dce049-ui/student-hub

G. Challenges Faced:

- Understanding the relationships between different tables.
- Defining correct primary and foreign keys.
- Maintaining proper database normalization.
- Handling database connection and SQL errors.

H. Learning Outcomes:

- Learned how to design and create a database schema.
- Understood primary keys and foreign keys.
- Learned the importance of database normalization.
- Learned how to connect an application with a database securely.
- Improved understanding of SQL queries and data management.

Faculty Remarks: **(Keep this blank for faculty)**